

ADBMS – Assignment 1

Histogram

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1 Database

The database used for the experiment is the Join Order Benchmark (2013 Snapshot). The table cardinalities were obtained using `COUNT(*)`. For this experiment, the primary-key column is recorded as `id` for all tables.

2 Q1 – PostgreSQL Histograms

The PostgreSQL histograms for the `id` and `title` columns of the `title` table were examined using `pg_stats`.

2.1 Q1.1 Corresponding SQL

The following query retrieves the statistics maintained by PostgreSQL:

```
SELECT
    attname ,
    null_frac ,
    n_distinct ,
    correlation ,
    histogram_bounds
FROM pg_stats
WHERE tablename = 'title'
    AND attname IN ('id', 'title');
```

The statistics obtained for the two columns were:

Column	Null fraction	n_distinct	Correlation	Histogram buckets
id	0.0	-1.0	0.99812114	762
title	0.0	139335.0	0.40523243	1890

Table 1: PostgreSQL statistics for `title.id` and `title.title`

For `id`, PostgreSQL reported 101 histogram boundary values, corresponding to 100 histogram buckets. For `title`, PostgreSQL reported 101 boundary values, corresponding to 100 histogram buckets.

2.2 Q1.2 Type of Histogram

PostgreSQL uses an equi-depth style histogram for these statistics. The histogram boundaries divide the ordered sample into approximately equal numbers of observations per bucket.

2.3 Q1.3 Number of Buckets

The number of buckets represented by the PostgreSQL statistics is:

- `id`: 100 buckets
- `title`: 100 buckets

2.4 Q1.4 Representation of the Textual Attribute

The `title` attribute is textual, but PostgreSQL can still order text values according to the database's comparison rules. Therefore, the values can be sorted and histogram boundaries can be selected from that ordering.

The histogram does not convert the title into a numerical value for the purpose of the histogram. Instead, the boundary values themselves are strings, and PostgreSQL uses its text comparison semantics when estimating predicates involving the column.

2.5 Q1.5 Histogram Bucket Boundaries

ID boundaries

The PostgreSQL `id` histogram contains 101 boundary values:

```
{114,24573,50079,74448,102250,129121,155133,181016,206149,
233863,260767,287742,314506,339108,362177,386919,411783,
435703,459964,485728,512026,535879,560447,587620,614036,
640736,663165,686493,710381,733376,759225,782723,806012,
829052,852028,877501,903058,927725,953127,978610,1004595,
1032084,1058076,1083746,1108234,1131865,1155911,1179238,
1205508,1234082,1259069,1284674,1310759,1335478,1360463,
1386675,1412692,1435435,1462459,1488257,1514089,1541398,
1568648,1593315,1617722,1642842,1665716,1690730,1717894,
1739710,1765639,1790886,1817383,1842017,1864212,1890074,
1915985,1943166,1967914,1997705,2023158,2051349,2076415,
2103407,2127836,2151710,2175304,2201304,2224706,2251068,
2274191,2297960,2324975,2350972,2378017,2401656,2427498,
2477404,2502294,2528164}
```

Title boundaries

The PostgreSQL `title` histogram contains 101 boundary values. The boundary values are strings ordered according to PostgreSQL's text comparison rules.

```
{''Allo 'Allo!',(#1.140),(#1.254),(#1.4825),(#1.80),
(#13.62),(#2.36),(#3.70),(#55.74),(#9.6),(1974-03-04),
(1990-01-27),(1996-07-06),(1999-05-28),(2001-01-17),
(2002-08-20),(2004-03-19),(2005-07-16),(2006-11-20),
(2008-02-07),(2009-09-08),(2011-02-26),(2012-03-27),
(2013-02-22),"30 Years of Emmerdale","A Lesson in Murder",
Abgeschminkt,"Akordet e Kosovës - populllore",
"An Autopsy of The Killing","Arrival of Rex",
"Bad Luck Be a Camper Tonight/Step Clam",
"Ben and Burman Find God","Black Bunbusters",
"Boy from Woy","Caffe Lena","Chamana Atha Guntha",
"Cities of Refuge",Cortège,"Dancing Arabs","Deadly Advice",
"Der Umweg",Dinty,"Dr. Hugo","Ein Bett für Drei",
"Empress of Ireland","Extremely Yours, Stryc-9",
"Final Approach","Frau Flinz","Genius of the Jet",
```

"Goode Lovin'", "Hard Times in the Country", Highlights,
 "Hotel Lolita 8", "If Lovin' You Is Wrong, I Don't Wanna Be Rat",
 "Internet Zombies!", "Jed and the Countess", Kadetten,
 "Kirameite, watashi no naka no watashi", "L'ombre d'un doute",
 "Lagda Ishq Ho Gaya",
 "Leila Shems: Everyone Is Invited to the Pants Party Except Me",
 "Little Big Head Man", "Love Thy Neighbor",
 "Männer soll man nicht alleine lassen", "Ménage à Trois",
 "Mob Wives Chicago", "Mud Show", Naujawan, "Nobody's Darling",
 "Oh Brother/Taz Babies", Otryv, Path-o-Prasad, Pivete,
 "Prince William Reveals How the Queen Helped Plan His Wedding",
 "Rafferty's Rise", Reunions, "Royal Cat Nap", "Sati Pingala",
 "Sentenza di morte", "Show Biz",
 "Snow White and the Juniper Tree", "Srdele, lignje i leptiri",
 "Summer Solstice Festival", "Tap Time", "The Annihilators",
 "The Case of the Horrifying Virus: Part 1", "The Erased",
 "The Haunted Planet", "The Living City",
 "The Nomination of Abraham Lincoln", "The Sandman",
 "The Tramp and the Tenner", "This Is My Past",
 "Top 2 Performance", Turnaround,
 "Une cicatrice derrière la tête", Vertauscht,
 "Wave Skiing 2.0 Jaws", "Who Killed Me?", "Y-2 Pappy",
 "Zzaap and the Word Master"}

2.6 Q1.6 Histogram Plots

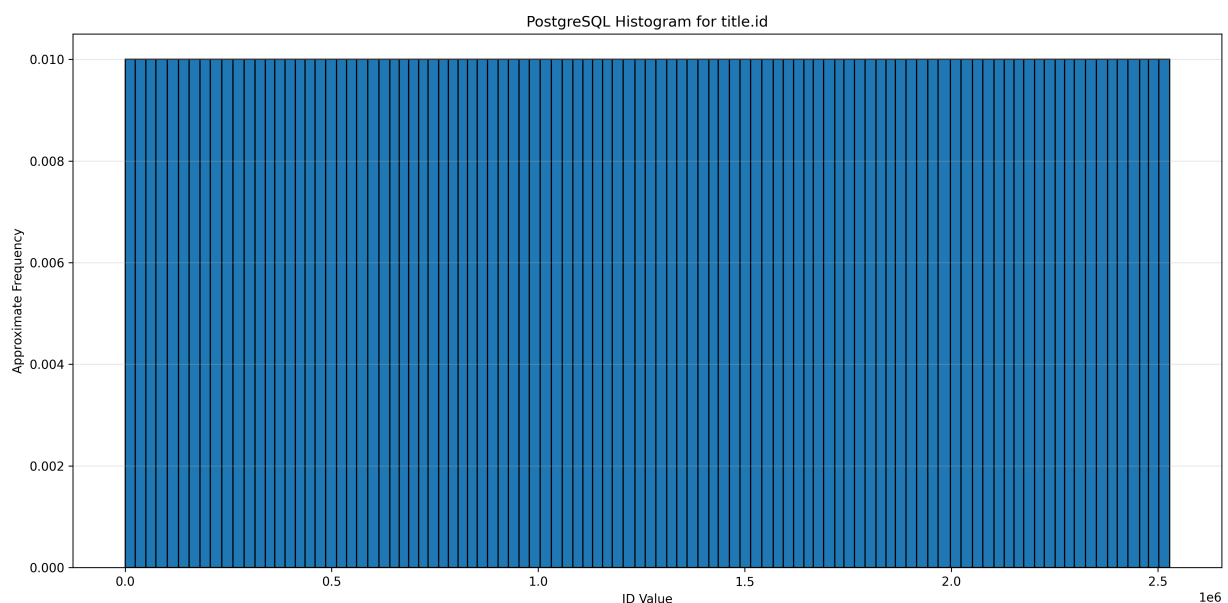
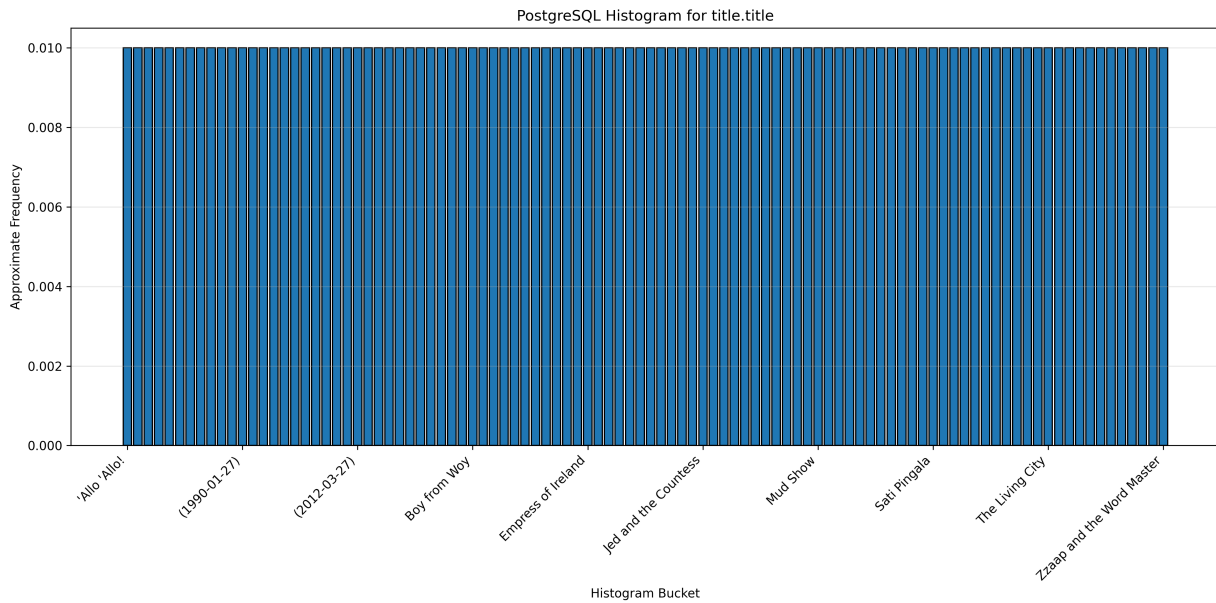


Figure 1: PostgreSQL histogram for `title.id`

Figure 2: PostgreSQL histogram for `title.title`

2.7 Q1.7 Maximum Selectivity Error

For a query Q , the selectivity error is

$$\text{Selectivity Error}(Q) = |S_{\text{estimated}}(Q) - S_{\text{actual}}(Q)|.$$

The maximum selectivity error is

$$E_{\max} = \max_Q |S_{\text{estimated}}(Q) - S_{\text{actual}}(Q)|.$$

For the Q1 experiment, the maximum selectivity errors obtained were:

$$E_{\max, \text{id}} = 0.004393, \quad E_{\max, \text{title}} = 0.064477$$

2.8 Q1.8 Selectivity Estimation for a Text Range Query

For a predicate such as

```
title < 'Race'
```

PostgreSQL determines the position of the string 'Race' in the ordered histogram. It identifies the bucket containing the value and estimates the fraction of values below the predicate value using the histogram information.

The estimate therefore depends on the histogram boundaries and the relative position of the query value within the appropriate bucket.

2.9 Q1.9 Estimated and Actual Selectivities

Query 1

```
SELECT *
FROM title
WHERE title.title < 'Race';
```

Estimated selectivity:

$$S_{\text{estimated}} = 0.758368$$

Actual selectivity:

$$S_{\text{actual}} = 0.759357$$

Therefore,

$$E = |0.758368 - 0.759357| = 0.000989.$$

Query 2

```
SELECT *  
FROM title  
WHERE title.id < 350000;
```

Estimated selectivity:

$$S_{\text{estimated}} = 0.134734$$

Actual selectivity:

$$S_{\text{actual}} = 0.138432$$

Therefore,

$$E = |0.134734 - 0.138432| = 0.003698.$$

3 Q2 – Optimal Serial Histogram

A serial optimal histogram was implemented using dynamic programming. The experiment uses:

- Number of buckets: **20**
- Sample sizes: **1000, 3000, 5000**
- Columns: **id, title**

Twenty buckets is within the required range of 10–50 buckets.

The samples were obtained from the PostgreSQL table using `TABLESAMPLE SYSTEM`.

3.1 Q2.1 Histograms

The Q2 plots use frequency on the X-axis and value-boundaries on the Y-axis, matching the orientation used for the serial histogram plots.

3.1.1 ID – 1000 samples

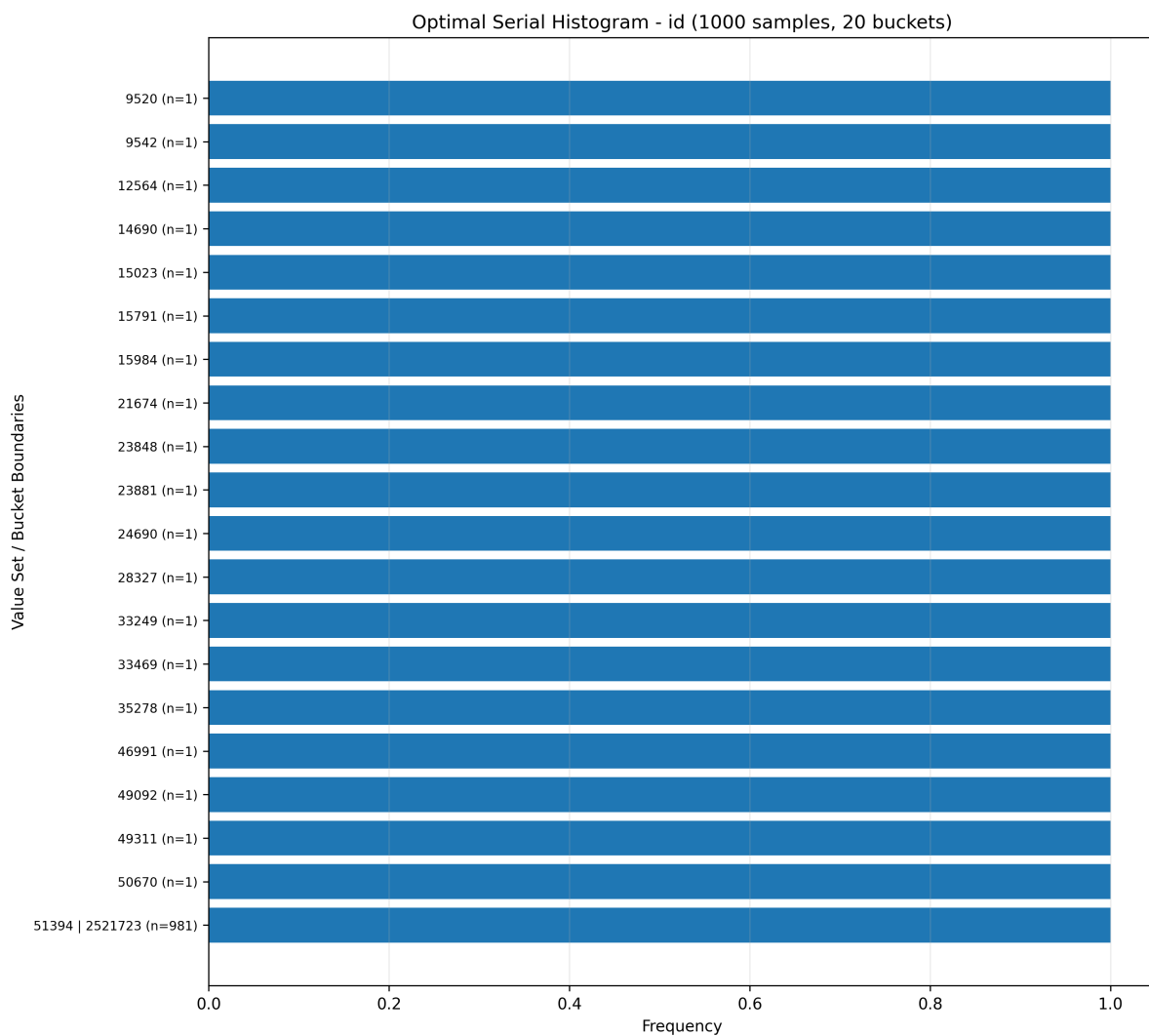


Figure 3: Optimal serial histogram for `id`, 1000 samples, 20 buckets

3.1.2 ID – 3000 samples

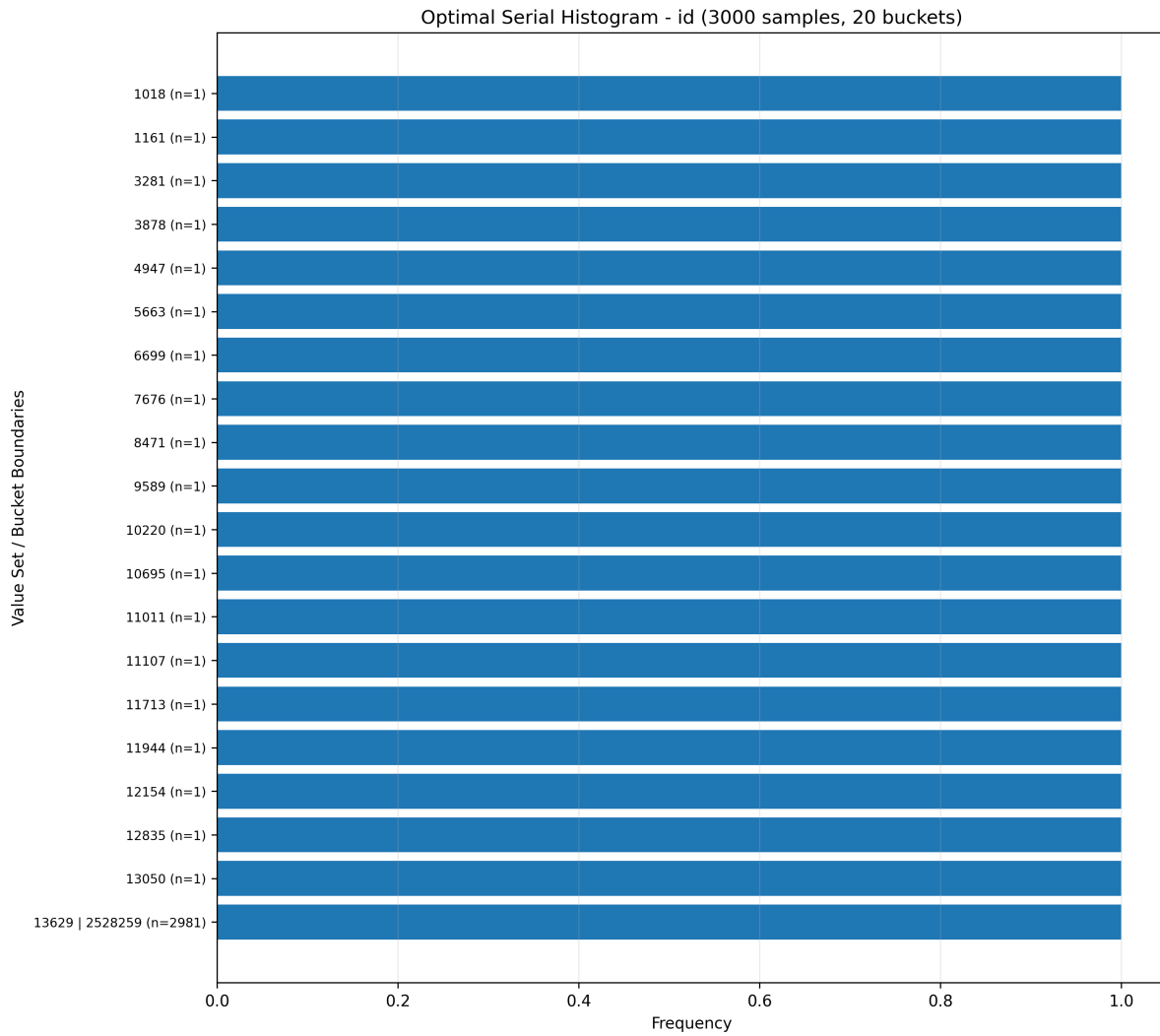


Figure 4: Optimal serial histogram for id, 3000 samples, 20 buckets

3.1.3 ID – 5000 samples

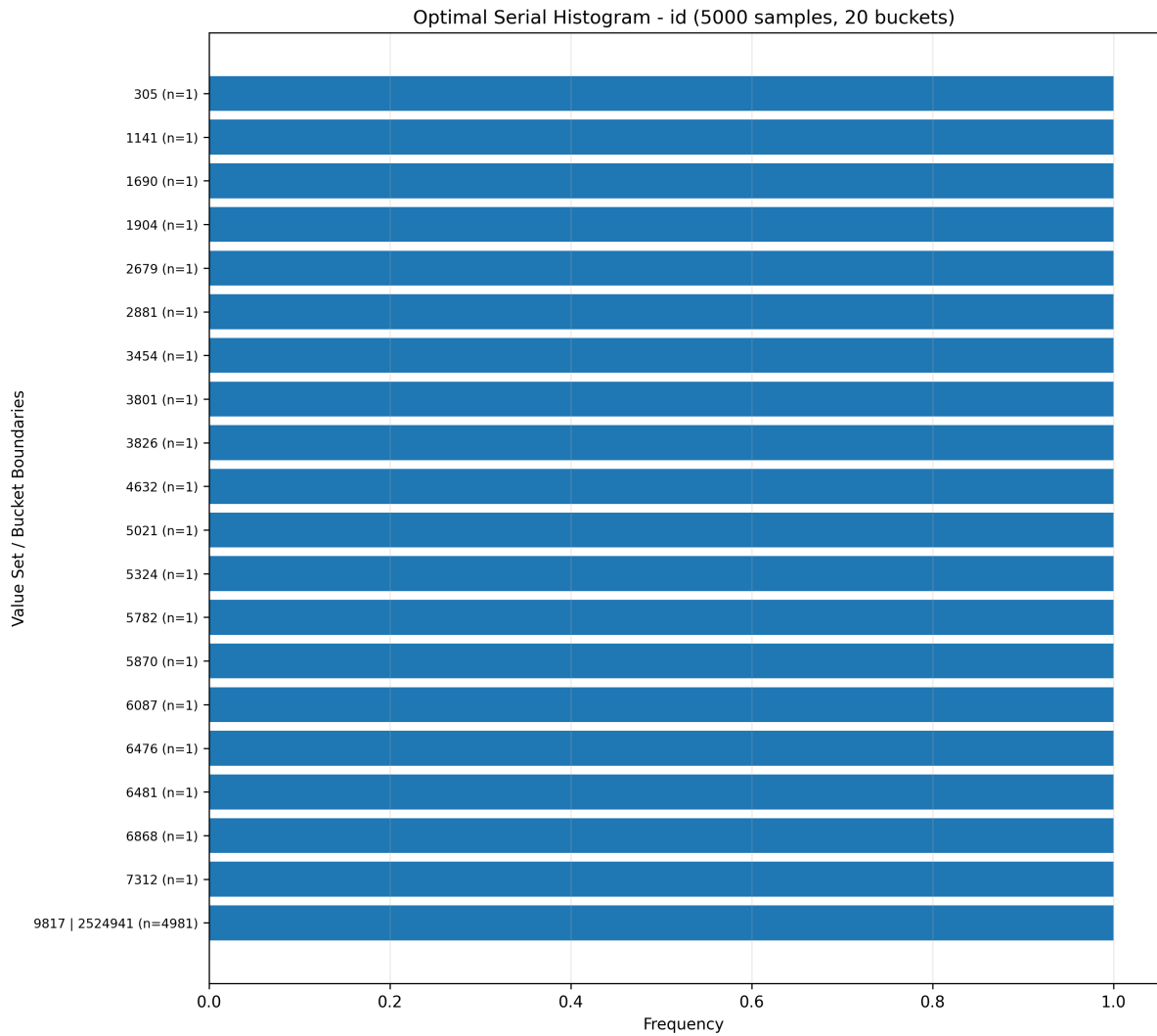
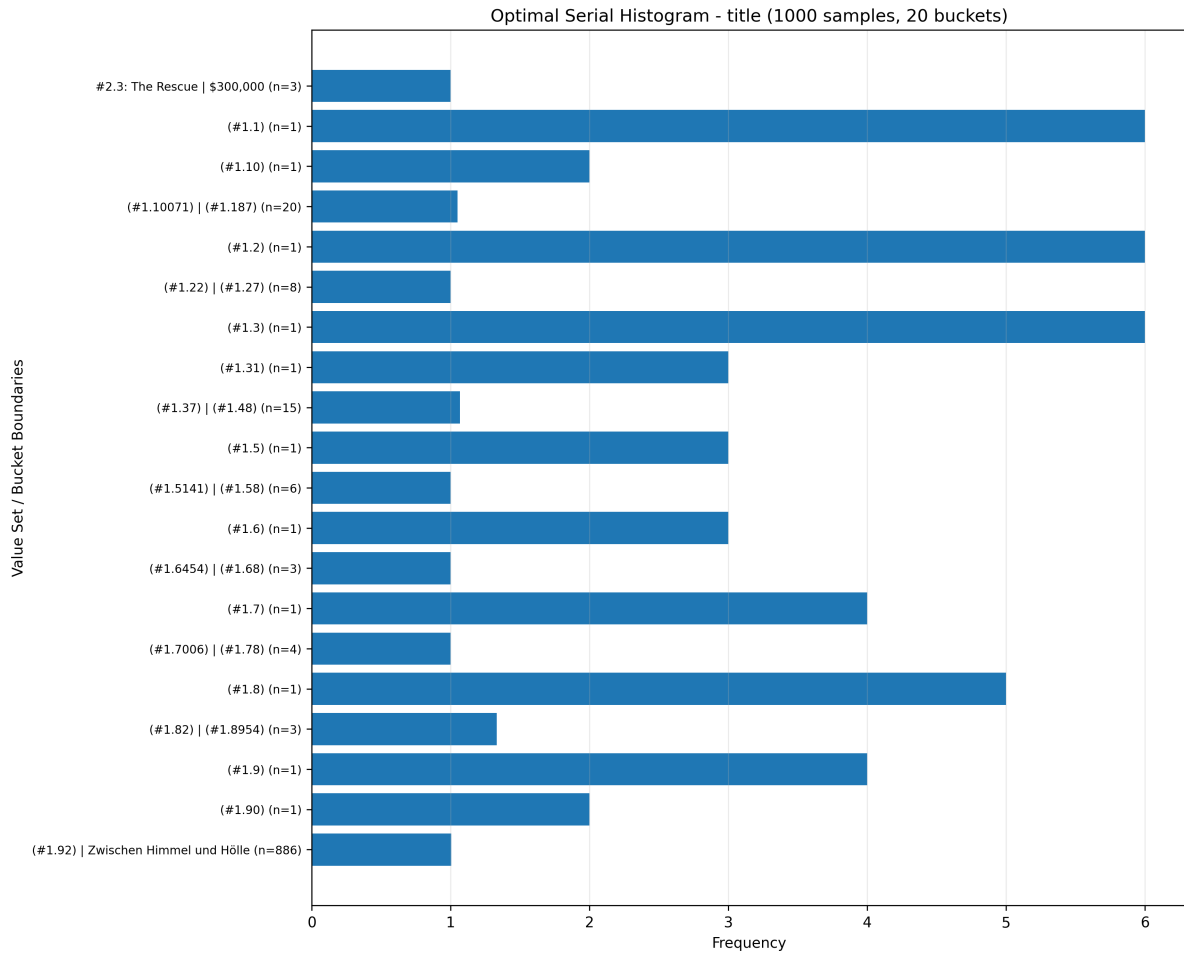


Figure 5: Optimal serial histogram for id, 5000 samples, 20 buckets

3.1.4 Title – 1000 samples

Figure 6: Optimal serial histogram for `title`, 1000 samples, 20 buckets

3.1.5 Title – 3000 samples

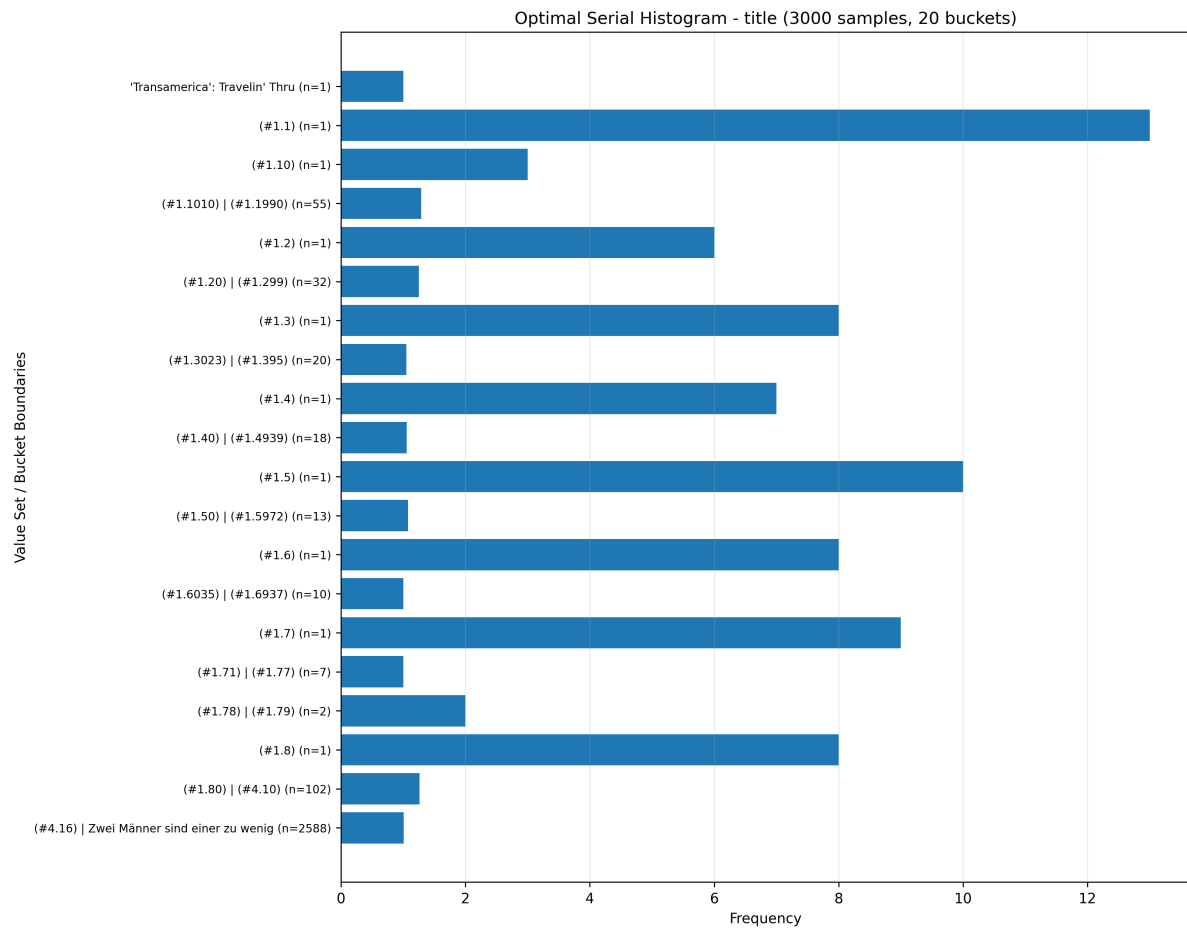


Figure 7: Optimal serial histogram for title, 3000 samples, 20 buckets

3.1.6 Title – 5000 samples

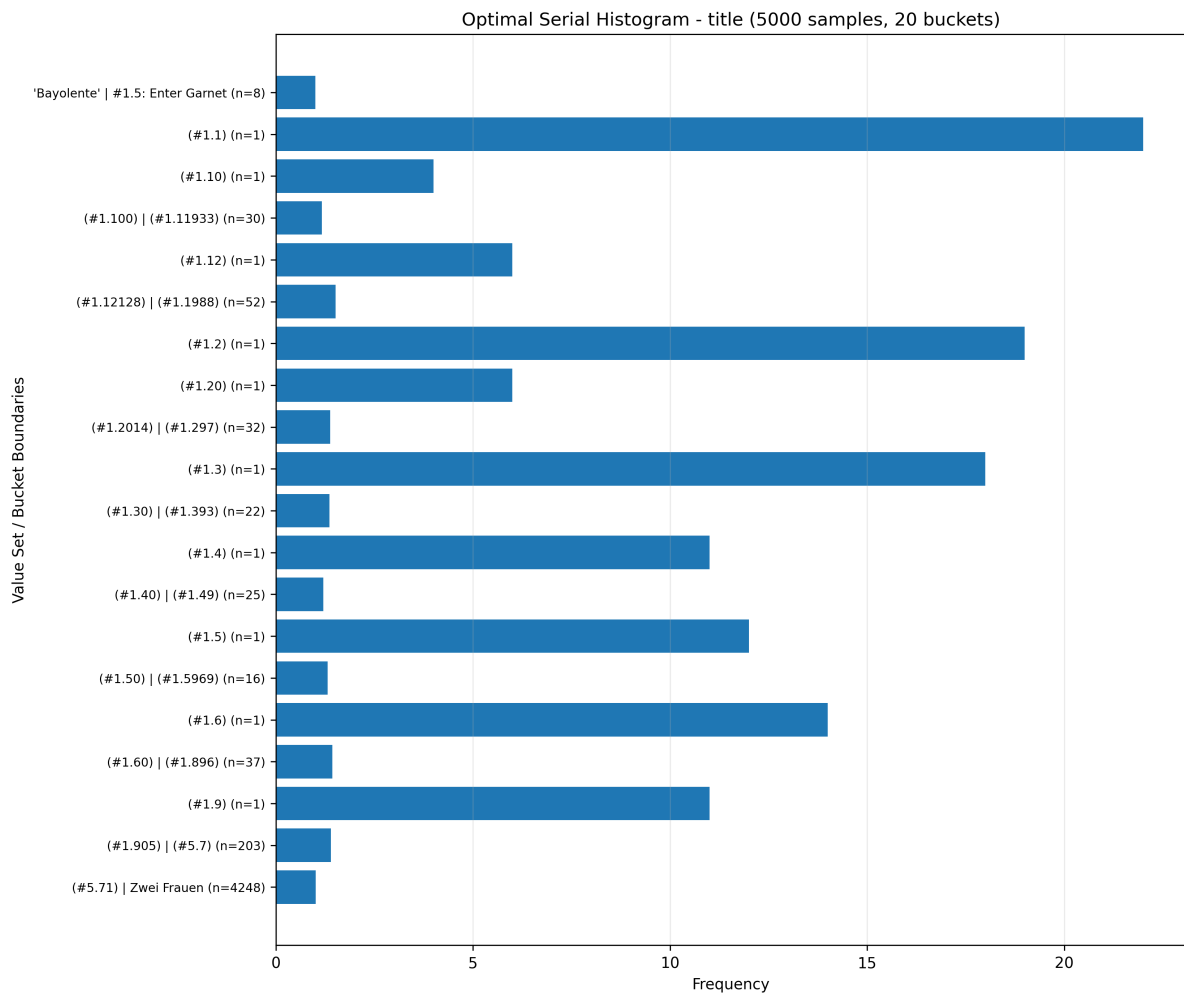


Figure 8: Optimal serial histogram for `title`, 5000 samples, 20 buckets

3.2 Q2.2 Histogram Construction Time

The final experimental run produced the following construction times:

Column	1000 samples	3000 samples	5000 samples
id	0.183802 s	0.745927 s	1.139123 s
title	0.351736 s	0.597718 s	1.051275 s

Table 2: Final histogram construction times

The full-table cardinality of the `title` table is

$$N = 2,528,312.$$

Using a linear fit between sample size and construction time, the extrapolated full-table construction times were:

Column	Extrapolated Full-table Time
id	603.810324 s = 10.06 min
title	442.305404 s = 7.37 min

Table 3: Extrapolated full-table histogram construction times

3.3 Q2.3 Frequency Bucket Boundaries

The optimal serial histogram partitions the ordered distinct values into 20 buckets. The boundaries below are taken directly from the final experimental run.

For each bucket, the reported lower and upper values define the range of distinct values assigned to that bucket.

ID – 1000 samples

Bucket	Lower Boundary	Upper Boundary	Distinct Values
1	9520	9520	1
2	9542	9542	1
3	12564	12564	1
4	14690	14690	1
5	15023	15023	1
6	15791	15791	1
7	15984	15984	1
8	21674	21674	1
9	23848	23848	1
10	23881	23881	1
11	24690	24690	1
12	28327	28327	1
13	33249	33249	1
14	33469	33469	1
15	35278	35278	1
16	46991	46991	1
17	49092	49092	1
18	49311	49311	1
19	50670	50670	1
20	51394	2521723	981

Table 4: Optimal ID histogram boundaries for 1000 samples

ID – 3000 samples

Bucket	Lower Boundary	Upper Boundary	Distinct Values
1	1018	1018	1
2	1161	1161	1
3	3281	3281	1
4	3878	3878	1
5	4947	4947	1
6	5663	5663	1
7	6699	6699	1
8	7676	7676	1
9	8471	8471	1
10	9589	9589	1
11	10220	10220	1
12	10695	10695	1
13	11011	11011	1
14	11107	11107	1
15	11713	11713	1
16	11944	11944	1
17	12154	12154	1
18	12835	12835	1
19	13050	13050	1
20	13629	2528259	2981

Table 5: Optimal ID histogram boundaries for 3000 samples

ID – 5000 samples

Bucket	Lower Boundary	Upper Boundary	Distinct Values
1	305	305	1
2	1141	1141	1
3	1690	1690	1
4	1904	1904	1
5	2679	2679	1
6	2881	2881	1
7	3454	3454	1
8	3801	3801	1
9	3826	3826	1
10	4632	4632	1
11	5021	5021	1
12	5324	5324	1
13	5782	5782	1
14	5870	5870	1
15	6087	6087	1
16	6476	6476	1
17	6481	6481	1
18	6868	6868	1
19	7312	7312	1
20	9817	2524941	4981

Table 6: Optimal ID histogram boundaries for 5000 samples

Title – 1000 samples

Bucket	Lower Boundary	Upper Boundary	Distinct Values
1	#2.3: The Rescue	\$300,000	3
2	(#1.1)	(#1.1)	1
3	(#1.10)	(#1.10)	1
4	(#1.10071)	(#1.187)	20
5	(#1.2)	(#1.2)	1
6	(#1.22)	(#1.27)	8
7	(#1.3)	(#1.3)	1
8	(#1.31)	(#1.31)	1
9	(#1.37)	(#1.48)	15
10	(#1.5)	(#1.5)	1
11	(#1.5141)	(#1.58)	6
12	(#1.6)	(#1.6)	1
13	(#1.6454)	(#1.68)	3
14	(#1.7)	(#1.7)	1
15	(#1.7006)	(#1.78)	4
16	(#1.8)	(#1.8)	1
17	(#1.82)	(#1.8954)	3
18	(#1.9)	(#1.9)	1
19	(#1.90)	(#1.90)	1
20	(#1.92)	Zwischen Himmel und Hölle	886

Table 7: Optimal title histogram boundaries for 1000 samples

Title – 3000 samples

Bucket	Lower Boundary	Upper Boundary	Distinct Values
1	'Transamerica': Thru	'Transamerica': Thru	1
2	(#1.1)	(#1.1)	1
3	(#1.10)	(#1.10)	1
4	(#1.1010)	(#1.1990)	55
5	(#1.2)	(#1.2)	1
6	(#1.20)	(#1.299)	32
7	(#1.3)	(#1.3)	1
8	(#1.3023)	(#1.395)	20
9	(#1.4)	(#1.4)	1
10	(#1.40)	(#1.4939)	18
11	(#1.5)	(#1.5)	1
12	(#1.50)	(#1.5972)	13
13	(#1.6)	(#1.6)	1
14	(#1.6035)	(#1.6937)	10
15	(#1.7)	(#1.7)	1
16	(#1.71)	(#1.77)	7
17	(#1.78)	(#1.79)	2
18	(#1.8)	(#1.8)	1
19	(#1.80)	(#4.10)	102
20	(#4.16)	Zwei Männer sind einer zu wenig	2588

Table 8: Optimal title histogram boundaries for 3000 samples

Title – 5000 samples

Bucket	Lower Boundary	Upper Boundary	Distinct Values
1	'Bayolente'	#1.5: Enter Garnet	8
2	(#1.1)	(#1.1)	1
3	(#1.10)	(#1.10)	1
4	(#1.100)	(#1.11933)	30
5	(#1.12)	(#1.12)	1
6	(#1.12128)	(#1.1988)	52
7	(#1.2)	(#1.2)	1
8	(#1.20)	(#1.20)	1
9	(#1.2014)	(#1.297)	32
10	(#1.3)	(#1.3)	1
11	(#1.30)	(#1.393)	22
12	(#1.4)	(#1.4)	1
13	(#1.40)	(#1.49)	25
14	(#1.5)	(#1.5)	1
15	(#1.50)	(#1.5969)	16
16	(#1.6)	(#1.6)	1
17	(#1.60)	(#1.896)	37
18	(#1.9)	(#1.9)	1
19	(#1.905)	(#5.7)	203
20	(#5.71)	Zwei Frauen	4248

Table 9: Optimal title histogram boundaries for 5000 samples

3.4 Q2.4 Maximum Selectivity Error

For each generated histogram, the maximum selectivity error was computed as

$$E_{\max} = \max_x |S_{\text{estimated}}(x) - S_{\text{actual}}(x)|.$$

The final run produced the following values:

Column	1000 samples	3000 samples	5000 samples
id	0.000000	0.000000	0.000000
title	0.000994	0.001664	0.001321

Table 10: Maximum selectivity errors from the final run

For the `id` column, the maximum selectivity error is zero for all three sample sizes because `id` is a primary key and every sampled value is unique.

For the `title` column, the maximum errors are small, ranging from 0.000994 to 0.001664.

The denominator used to compute selectivity is the number of rows in the sample used to construct and evaluate the histogram.

3.5 Q2.5 Effect of Sample Size

The experiment was repeated for 1000, 3000, and 5000 samples while keeping the number of buckets fixed at 20.

Measure	1000	3000	5000
id distinct values	1000	3000	5000
title distinct values	959	2857	4683
id build time (s)	0.183802	0.745927	1.139123
title build time (s)	0.351736	0.597718	1.051275
id objective	0.000000	0.000000	0.000000
title objective	7.521783	151.793351	458.231716
id max error	0.000000	0.000000	0.000000
title max error	0.000994	0.001664	0.001321

Table 11: Effect of sample size on the optimal serial histogram

The number of distinct sampled `id` values is equal to the sample size because `id` is unique. For `title`, the number of distinct values is lower because multiple rows can have the same title.

3.6 Q2.6 Optimal Histogram Objective

The optimal serial histogram minimizes

$$\sum_{i=1}^B n_i V_i$$

where B is the number of buckets, n_i is the number of sampled values in bucket i , and V_i is the variance of the frequencies of the distinct values contained in bucket i .

The final objective values were:

Column	1000	3000	5000
id	0.000000	0.000000	0.000000
title	7.521783	151.793351	458.231716

Table 12: Optimal serial histogram objective values

For `id`, every value occurs exactly once in the sample. Consequently, the frequency variance in every bucket is zero and the objective is exactly zero.

For `title`, repeated values occur with different frequencies, so the objective is non-zero.

3.7 Q2.7 Full-table Time Extrapolation

A linear model was fitted between sample size and histogram construction time.

For the `id` column, the fitted model was

$$T_{\text{id}}(n) = 0.000238830175n - 0.026873024957.$$

Using

$$n = 2,528,312,$$

the extrapolated construction time is

$$T_{\text{id}} = 603.810324 \text{ seconds} \approx 10.06 \text{ minutes}.$$

For the `title` column, the fitted model was

$$T_{\text{title}}(n) = 0.000174884725n + 0.142255291789.$$

Therefore,

$$T_{\text{title}} = 442.305404 \text{ seconds} \approx 7.37 \text{ minutes}.$$

These are extrapolated values rather than directly measured full-table construction times.

3.8 Q2.8 Sample Size vs. Construction Time

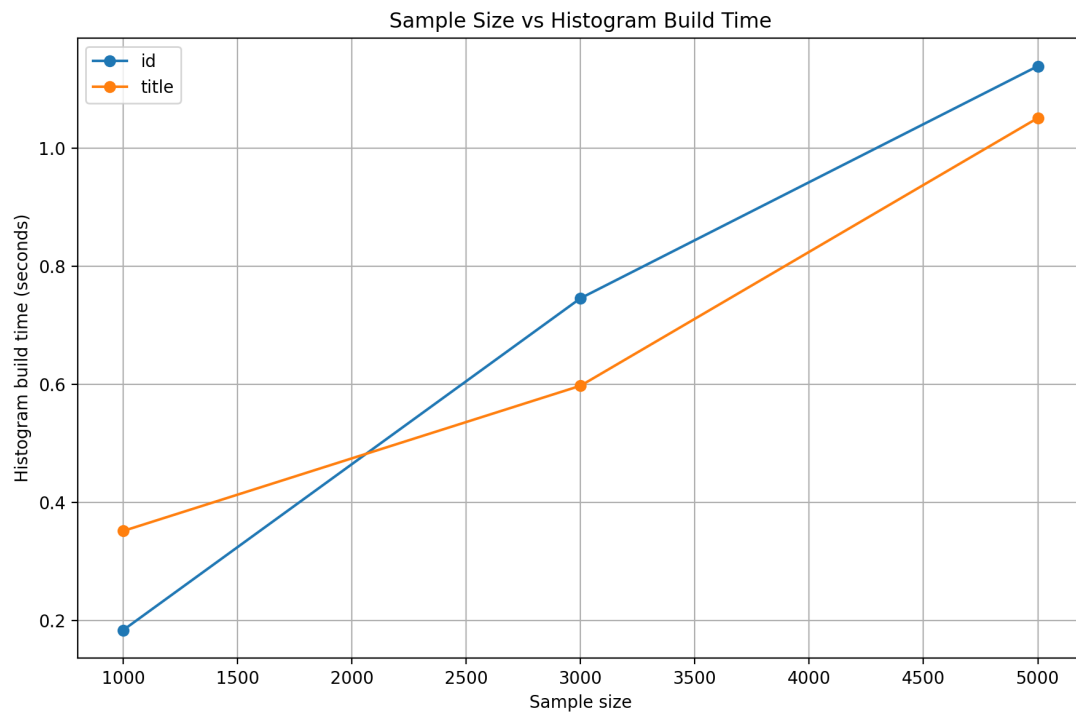


Figure 9: Sample size versus histogram construction time

The construction time generally increases as the sample size increases. This is expected because the optimization algorithm has more sampled values to process.

3.9 Q2.9 Sample Size vs. Maximum Selectivity Error

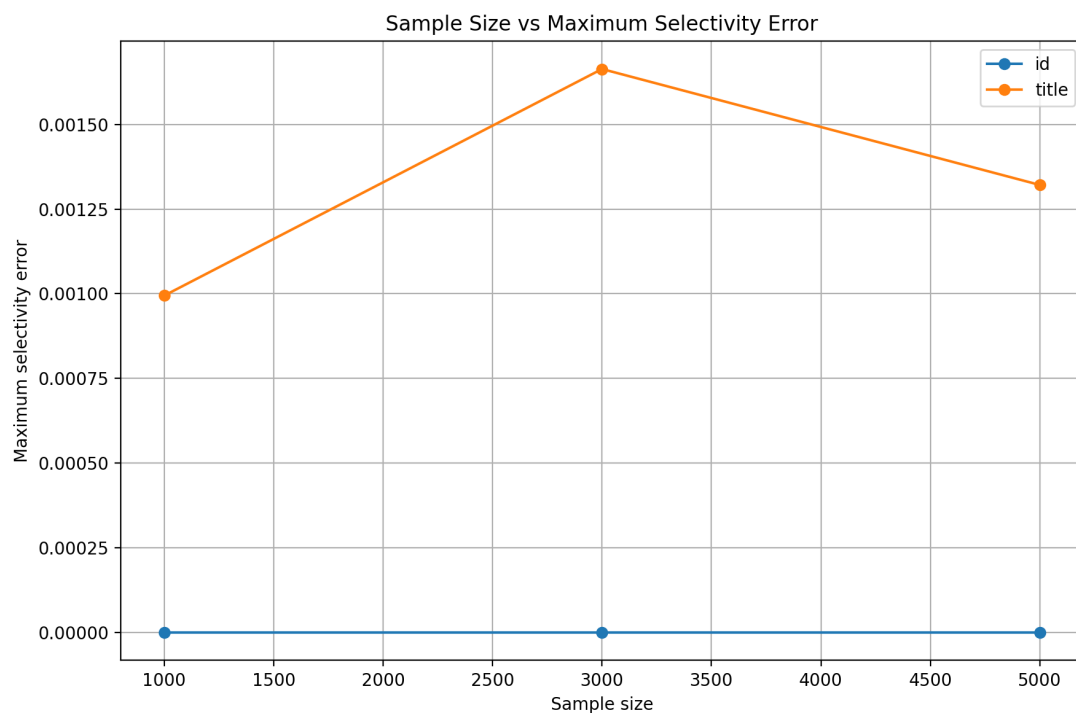


Figure 10: Sample size versus maximum selectivity error

The maximum selectivity error is not necessarily strictly monotonic with sample size because the samples are randomly selected. Increasing the sample size provides more information about the underlying distribution, but the exact measured error can still vary between samples.

3.10 Q2.10 Effect of the Number of Buckets

If fewer buckets are used, each bucket represents a larger portion of the value domain. This reduces the amount of information retained by the histogram and can increase selectivity estimation error.

If more buckets are used, the distribution is represented in greater detail. This can reduce selectivity error, especially when the data distribution is irregular. However, more buckets also require more boundary information and may increase histogram construction cost.

Therefore, the number of buckets represents a trade-off between estimation accuracy and histogram construction/storage cost.

4 Final Results Summary

Column	Samples	Time (s)	Buckets	Objective	Max Error
id	1000	0.183802	20	0.000000	0.000000
title	1000	0.351736	20	7.521783	0.000994
id	3000	0.745927	20	0.000000	0.000000
title	3000	0.597718	20	151.793351	0.001664
id	5000	1.139123	20	0.000000	0.000000
title	5000	1.051275	20	458.231716	0.001321

Table 13: Final Q2 experimental results